Marco Vinicio Alban-Paccha, MEng, PhD, FHEA, MIEEE

Email: marco.albanpaccha@ucd.ie | marcoalban.com | LinkedIn | Google Scholar | ORCID

Engineer, scientist, and educator working on creating technologies that seamlessly connect the physical and digital worlds. Working across every layer: from material foundations to intelligent systems, my vision is to lead innovative research at the intersection of engineering, healthcare, and clinical translation.

ACADEMIC APPOINTMENTS

BDIC Lecturer/Assistant Professor in Electronic Engineering **2025 – present**
School of Electrical and Electronic Engineering, University College Dublin, Dublin, Ireland

Postdoctoral Research Associate **2022 – 2025**
Department of Engineering, Department of Medicine, University of Cambridge, UK

Associate Lecturer, The Open University, Milton Keynes, UK **2022 – 2025**
Faculty of Science, Technology, Engineering and Mathematics

Postdoctoral By-Fellow, Churchill College, Cambridge, UK **2023 – 2025**

Lecturer, Universidad de las Américas – UDLA, Ecuador **2022**
Faculty of Online Studies

EDUCATION

PhD in Electrical Engineering **2018 – 2022**
Integrated Organic Electronics Laboratory, Korea Advanced Institute of Science and Technology – KAIST, Daejeon, South Korea
Title: *Applications of Flexible Dry Electrodes in Biopotential-based Real-time Cardiac Monitoring*
Supervisor: Prof Seunghyup Yoo

MEng in Micro/Nano Systems **2016 – 2018**
Display and Nanosystems Laboratory, Korea University, Seoul, South Korea
Title: *Optimization of Electron Injection in Organic Light Emitting Diode (OLED) Devices Using Alkali Metal Compounds*
Supervisor: Prof Byeong-Kwon Ju

B2 Certificate in Korean Language **2014 – 2016**
Pai Chai University, Daejeon, South Korea
As part of the requisites for the Korean Government Scholarship Program

Engineering Degree in Mechatronics Engineering **2006 – 2013**
University of the Armed Forces – ESPE, Sangolquí, Ecuador
Title: *Design and Implementation of a SCADA System for Remote Operation of Manufacturing Stations via Internet2*
Supervisor: Dr Alejandro Chacón

PUBLICATIONS

1. Alban-Paccha, M.V.*, Shenker, N., Teran-Perez, J., Horne, A.W., Malliaras, G.G., Woods, C.G., and the ADVANTAGE Consortium (2026). *ADVANTAGE: Advanced Discovery of Visceral Analgesics by Neuroimmune Targets and the Genetics of Extreme human phenotype, a study protocol*. PLOS ONE, https://doi.org/10.1371/journal.pone.0350169
2. Kaya, O., Athanassopoulou, N., Malliaras, G.G., and Alban-Paccha, M.V.* (2026). *Differentiating Physical and Psychological Stress Using Wearable Physiological Signals and Salivary Cortisol*. arXiv Preprint, https://doi.org/10.48550/arXiv.2604.12671
3. Sills, V.A.*, Rennie, K.L., Rzechorzek, N.M., Watson, C.J., Scott, S., Langford, J., Alderton, W., Li, N., Mascolo, C., Martinez-Hernandez, V., Antoniou, A., Alban-Paccha, M.V., Shreves, A.H., Flewitt, A.J., and Fitzgerald, R. (2025). *Can Wearable Technologies Assist in the Earlier Detection of Cancer?* Wellcome Open Research, https://doi.org/10.12688/wellcomeopenres.24402.1

4. Tao, X., Carnicer-Lombarte, A., Dominguez-Alfaro, A., Gatecliff, L., Zhang, J., Bidinger, S., Keene, S.T., El Hadwe, S., Dong, C., Boys, A.J., Slaughter, C., Ruiz-Mateos Serrano, R., Chovas, J., Alban-Paccha, M.V., Barone, D., Kar-Narayan, S., and Malliaras, G.G.* (2025). *Cleanroom-Free Toolkit for Integrating Submicron-Resolution Bioelectronics on Flexibles*. Small, https://doi.org/10.1002/smll.202411979
5. Serrano, R. R-M., Aguzin, A., Mitoudi-Vagourdi, E., Tao, X., Naegele, T., Jin, A., Lopez-Larrea, N., Picchio, M.L., Alban-Paccha, M.V., Minari, R.J., Mecerreyes, D., Dominguez-Alfaro, A.*, and Malliaras, G.G.* (2024). *3D Printed PEDOT:PSS-based Conducting and Patternable Eutectogel Electrodes for Machine Learning on Textiles*. Biomaterials, https://doi.org/10.1016/j.biomaterials.2024.122624
6. Alban, M.V., Lee, H., Moon, H., and Yoo, S.* (2021). *Micromolding Fabrication of Biocompatible Dry Micro-Pyramid Array Electrodes for Wearable Biopotential Monitoring*. IOP Flexible and Printed Electronics, https://doi.org/10.1088/2058-8585/ac3561
7. Lee, H., Lee, W., Lee, H., Kim, S., Alban, M.V., Song, J., Kim, T., Lee, S., and Yoo, S.* (2021). *Organic–Inorganic Hybrid Approach to Pulse Oximetry Sensors with Reliability and Low Power Consumption*. ACS Photonics, https://doi.org/10.1021/acsphotonics.1c01161

Edited Books

1. **Alban-Paccha, M.V.**, Han, S., Keene, S., and Gatecliff, L. (Editors). (Under Contract). *Organic Electrochemical Transistors For Wearable And Implantable Sensing: From Fundamentals To Clinical Translation*. Elsevier.

FUNDING

Principal Investigator

1. School-funded Internal Award (Alban-Paccha). Granted €110,000.00 to support doctoral students' university fees, stipend and research expenses from the Beijing-Dublin International College from 09/2026 to 08/2030.
2. Churchill College By-Fellowship (Alban-Paccha). Granted £3,000.00 for research expenses by Churchill College, Cambridge from 02/2023 to 02/2026.

Co-Investigator

1. CAPE Grand Challenge – Systems and Devices for Healthcare 2023 (PI: Malliaras, Co-I: Alban-Paccha). Granted £50,000.00 for salary and research expenses by Haleon from 12/2023 to 05/2024. Wrote proposal and held multiple meetings with company representatives.

Senior Personnel

1. MICA ADVANTAGE Visceral Pain Consortium: Advanced Discovery of Visceral Analgesics via Neuroimmune Targets and the Genetics of Extreme human phenotype (PI: Woods, St. John Smith). £4,101,154.00 by the Medical Research Council as part of UK Research and Innovation from 06/2022 to 05/2026. Data Manager and Lead of the wearable electronics and mobile apps arm of the study.

Scholarships

1. Attachable Photo Therapeutics Centre for e-Healthcare (PI: Yoo). Granted tuition, salary and research expenses by the Ministry of Science and ICT of Korea from 03/2018 to 02/2022.
2. Korean Government Scholarship Program, now known as the Global Korea Scholarship (Alban-Paccha). Granted tuition and student stipend by the Ministry of Education of Korea from 09/2014 to 02/2018.

PRESENTATIONS AND OUTREACH

Conference Presentations

1. Alban-Paccha, M.V., Gatecliff, L., Lee, C., Jang, S., Slaughter, C., Kissovsky, S., Keene, S.T., Han, S., and Malliaras, G.G. *Multimodal Transistorized Wearable Electrochemical Sensor Platform for Ion and Enzyme Analysis*. ZJU-Cambridge Youth Scientist Forum, Zhejiang University, Hangzhou, China, 29 August 2025
2. Alban-Paccha, M.V., Shenker, N., Woods, C.G., and Malliaras, G.G. *Predicting Pain Flares from Wearable Sensor Data and Patient Reports: Initial Insights from the ADVANTAGE Study*. Oral presentation at the 18th International Symposium on Flexible Organic Electronics ISFOE25, Thessaloniki, Greece, 8 July 2025.
3. Alban-Paccha, M.V.‡, Cullen, K.‡, Woods, C.G., Malliaras, G.G., and Shenker, N. *Visually Understanding Chronic Visceral Pain: Results from the ADVANTAGE UK National Survey on Visceral Pain*. Poster presentation at the 2025 APDP Annual Conference, Newport, Wales, 3 June 2025.

4. Alban-Paccha, M.V., Teran-Perez, J., Shenker, N., Woods, C.G., and Malliaras, G.G. *Wearable Sensors and Mobile App Data for the Modelling and Classification of Visceral Pain*. Oral presentation at the 17th International Symposium on Flexible Organic Electronics ISFOE24, Thessaloniki, Greece, 4 July 2024.
5. Alban-Paccha, M.V., Teran-Perez, J., Shenker, N., Woods, C.G., and Malliaras, G.G. *Wearable Sensors and Mobile App Data for the Modelling and Classification of Visceral Pain*. Oral presentation at the 'Can Cancer be Detected Earlier by Employing Wearable Technologies?' symposium organised by the Early Cancer Institute and Precision Health Initiative, Cambridge, UK, 20 October 2023.
6. Alban, M.V., Lee, H., Moon, H., and Yoo, S. *Biocompatible Microneedle Array Dry Electrodes for Bioelectric Potentials Measurement in Organic-Electronic Wearable Health Monitoring Applications*. Best Poster Award Nominee at MRS Fall 2019, Boston, USA, 4 December 2019.
7. Alban, M.V., Lee, H., Moon, H., and Yoo, S. *Flexible and Fully Biocompatible Microneedle Array Dry Electrodes for Bio Potentials Measurement in Organic Electronic Wearable Healthcare Applications*. Poster presentation delivered at Electronic Materials and Nano Technology for Green Environment ENGE 2018, Jeju, Korea, 19 November 2018
8. Alban, M.V., Choi, J., Jung, S.G., Shim, Y.S., Park, Y.W., and Ju, B.K. *Comparative study of different alkali metal compounds as efficient electron injection materials in OLED devices*. Best Poster Award at the Workshop on Photophysics and Nanomaterials WONPHYS 2017, Varadero, Cuba, 27 September 2017.

Invited Presentations

1. *Wearable OECT Technology for Multimodal Electrochemical Sensing*. Korea Advanced Institute of Science and Technology – KAIST, Daejeon, Korea, June 2026
2. *Multimodal Transistorized Wearable Electrochemical Sensor Platform for Ion and Enzyme Analysis*. Aristotle University of Thessaloniki, Thessaloniki, Greece, July 2025
3. *Wearable Sensors and Mobile App Data for the Modelling and Classification of Visceral Pain Flares*. ADVANTAGE General Annual Meeting, University of Edinburgh, Edinburgh, UK, April 2024
4. *Innovative Wearables and Mobile Apps for Managing Visceral Pain: Insights from the ADVANTAGE Consortium*. Addenbrooke's Hospital, Cambridge, UK, July 2023
5. *Cutting-Edge Wearables and Mobile Applications for Visceral Pain Management: Perspectives from the ADVANTAGE Consortium*. Korea Advanced Institute of Science and Technology – KAIST, Daejeon, Korea, June 2023
6. *Revolutionising Visceral Pain Management: Advances in Wearable Technology and Mobile Applications from the ADVANTAGE Consortium*. Pohang University of Science and Technology – POSTECH, Pohang, Korea, June 2023
7. *Academic Achievements and Research Outcomes Following the Korean Government Scholarship*. Embassy of the Republic of Korea in Ecuador, Quito, Ecuador, July 2022
8. *Enhancing Educational Excellence: The Impact of Korea-Trained Researchers' Skill Transfer*. SWIFT Talks, San Francisco de Quito University – USFQ, Quito, Ecuador, February 2020
9. *Organic Electronics Masterclass*. Armed Forces University – ESPE, Sangolquí, Ecuador, February 2020

TEACHING AND SUPERVISION

Courses Taught

1. *EEEN3007J: Embedded Systems*, University College Dublin — 2026 – Present
2. *EEEN3012J: Embedded Systems and Software*, University College Dublin — 2026 – Present
3. *EEEN3005J: Communication Theory*, University College Dublin — 2026 – Present
4. *T366: Nanoscale Engineering*, The Open University — 2022 – 2025
5. *Intelligent Systems*, Universidad de las Américas – UDLA — 2022

Final Year Project Supervision

1. *IoT Final Year Project*, University College Dublin — 2026
2. *T452: The Engineering Project*, The Open University — 2023–2025

Laboratory Supervision / Courses Supported

1. *Part IIB Bioelectronics Lab*, University of Cambridge — 2024 – 2025
2. *Part IA Computing Supervisions*, Homerton College, University of Cambridge — 2022 – 2025
3. *Part IA Computing Help Desk*, University of Cambridge — 2023 – 2024
4. *Computer Integrated Manufacture Lab*, University of the Armed Forces – ESPE — 2012 – 2013

5. *Instrumentation Lab*, University of the Armed Forces – ESPE 2011

<u>Supervision of Graduate Students</u>

1. Roman Percht, doctoral candidate in Electrical and Electronic Engineering, University College Dublin. Title: *TBD (*2026 – present)
2. Luke Gatecliff, doctoral candidate in Engineering, University of Cambridge. Title: *OECT-based Ion Sensors for Athletics and Healthcare* (2022 – 2026)
3. Christopher Slaughter, doctoral candidate in Engineering, University of Cambridge. Title: *Biopotential Gastric Movement Measurement Platform* (2023 – 2025)
4. Kieran Cullen, doctoral candidate in Medicine, University of Cambridge. Title: *Analysis of Body-Map dataset from the ADVANTAGE Pain Survey* (2024 – 2025)
5. Bartosz Zygowski, undergraduate summer Intern in Engineering, University of Cambridge. Title: *Optimisation of Integrated Microfluidic Devices* (2025)
6. Gemma Jacobson, MEng (Hons), University of Cambridge. Title: *Cuffless Biopotential-Based Blood Pressure Estimation* (2023 – 2024)
7. Cristiano Bortolotti, visitor from the Polytechnic University of Milan, Italy (2025)
8. Seungjin Chai, visitor from the Pohang University of Science and Technology – POSTECH, Korea (2023)

ACADEMIC SERVICE

- Reviewer, IEEE Sensors Letters, Institute of Electrical and Electronics Engineers (2026 – present)
- Reviewer, IEEE Biosensors Conference, Institute of Electrical and Electronics Engineers (2026 – present)
- Reviewer, Materials Research Express, Institute of Physics (2026 – present)
- Reviewer, Flexible and Printed Electronics, Institute of Physics (2023 – present)
- Reviewer, Nanotechnology, Institute of Physics (2023 – present)
- Administrator and Planner, PPI for Neurotech Initiative, Cambridge, UK (2024 – 2025)
- Reviewer, IEEE International Conference on Flexible and Printable Sensors and Systems, Institute of Electrical and Electronics Engineers (2023 – 2024)
- Council Head, Electrical Engineering International Students Council, KAIST, Daejeon, Korea (2019 – 2020).
- Co-Founder and President, Ecuadorian Residents in Korea Association, Seoul, Korea (2017 – 2018). Collaborated on legal framework recognition by the Korean Government and represented the Ecuadorian community in diverse events.
- Vice-President, School of Engineering International Students Group, Korea University, Seoul, Korea (2017 – 2018)